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AI for Bharat Hackathon

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Team Name : KrishiMitra-AI

Team Leader Name : Sagar Tate

Problem Statement : Build an AI-powered solution that supports rural ecosystems, sustainability, and resource-efficient systems.

Brief About the Idea

KrishiMitra-AI

An AI-powered Rural Decision Intelligence Platform designed to empower smallholder farmers with:

- Market intelligence
- Climate risk insights
- Sustainability analytics
- Responsible multilingual AI assistance

KrishiMitra-AI bridges the information gap in rural ecosystems by providing explainable, safe, and scalable AI support tailored for low-bandwidth and regional language users.

Problem We Are Solving

Smallholder farmers face:

- Lack of transparent market price information
- Uncertain weather and climate risks
- Limited advisory support
- Inefficient water and input usage
- Language and connectivity barriers

This leads to:

- Poor pricing decisions
- Income instability
- Resource wastage
- Increased vulnerability to climate change

How Is Our Solution Different?

- Decision Intelligence, not just Chatbot
  - Net-profit-based market recommendations
  - Transport cost estimation & volatility-aware forecasts
- Explainable AI
  - Top contributing factors, confidence scores, citation-backed responses
- Responsible AI by Design
  - Chemical dosage refusal, prompt injection protection, uncertainty handling
- Sustainability Built-In
  - Water efficiency benchmarking, climate risk index, resource optimization
- Offline-First & Multilingual
  - Progressive Web App, low-bandwidth mode, regional language support

USP (Unique Selling Proposition)

KrishiMitra-AI is the first integrated platform that combines:

- AI Copilot
- Market Forecasting
- Sustainability Analytics
- Climate Risk Alerts
- Crop Disease Vision AI

All within a multi-tenant, scalable SaaS architecture designed specifically for rural India.

Key Features

Market Intelligence
Price comparison across markets
Profit-based ranking
Forecast with confidence intervals
Price alerts

Climate Risk Intelligence
7-day risk index
Heat, drought, rainfall alerts
Preventive guidance

Sustainability Analytics
Water efficiency rating
Input cost tracking
Benchmark comparison

AI Assistant
Multilingual conversational support
Citation-based responses
Safety guardrails

Crop Disease Detection
Image classification
Confidence threshold
Safe treatment suggestions

Offline-First & Multilingual
Progressive Web App
Low-bandwidth mode
5 Indian languages

Process Flow Diagram

User Onboarding
Select Language → Create Farm Profile

↓

User Action
Ask AI / Upload Image / View Market Data

↓

System Processing
RAG Retrieval I Forecast I Risk Analysis I Disease Classification

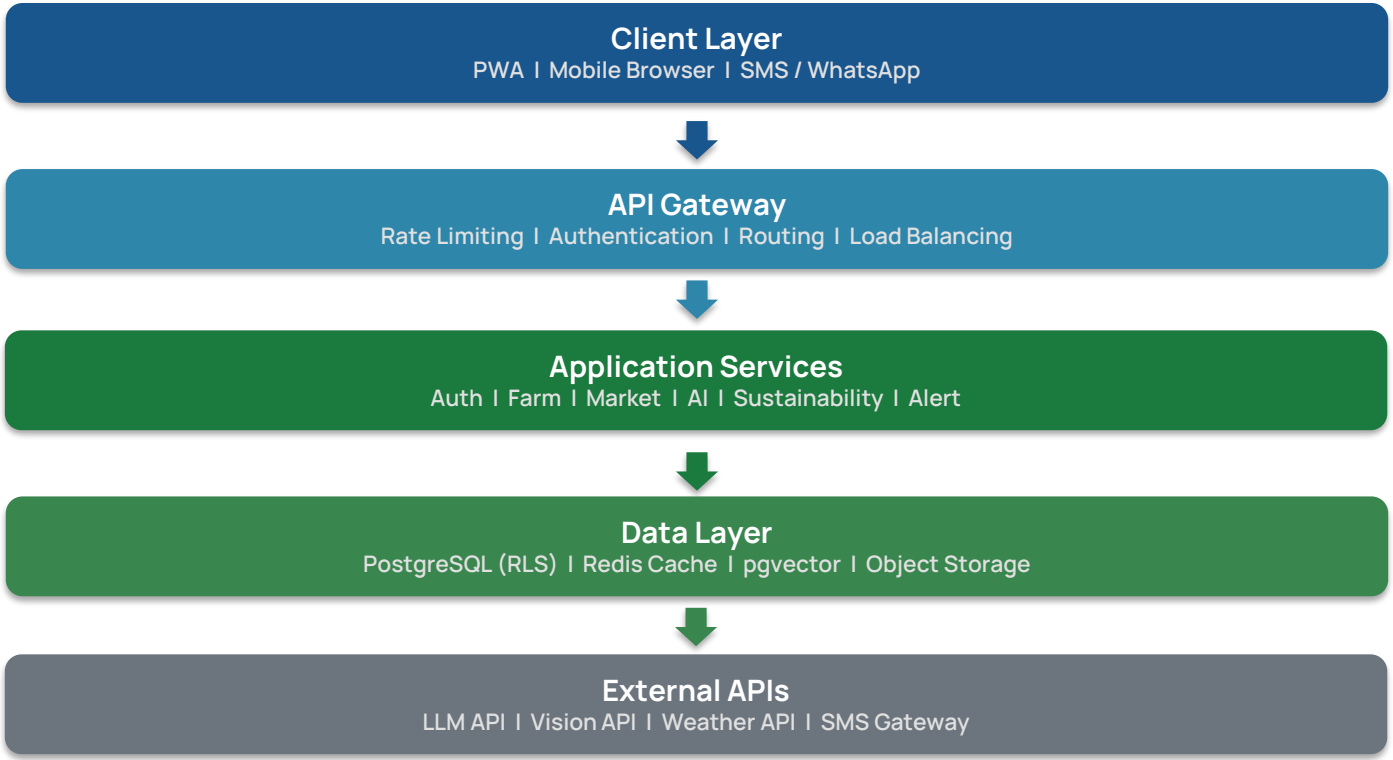
↓

Response Delivered
Explanation + Confidence Score + Citation + Disclaimer

↓

Alert Setup
Notifications via App / SMS

Architecture Diagram



Technologies Used

- Frontend
  - React + TypeScript (PWA)
- Backend
  - Node.js + TypeScript, PostgreSQL (RLS), Redis, RabbitMQ
- AI / ML
  - LLM (OpenAI / Azure OpenAI), RAG (pgvector)
  - ResNet / EfficientNet (Disease Classification), Speech APIs
- Cloud & Infrastructure
  - AWS / Azure, S3 Object Storage, CDN
- Testing
  - Unit + Property-Based Testing, OWASP Security Testing

Estimated Implementation Cost

MVP (1–5 tenants, 1,000 users)

- Cloud hosting: ~$150 – $300 / month
- LLM API usage: ~$200 / month (controlled usage)
- Storage & CDN: ~$50 / month

Total Estimated MVP Cost: ~$400 – $600 / month

Production cost scales based on user growth and AI usage.

Why This Matters

KrishiMitra-AI empowers rural communities with:

- Data-driven confidence
- Sustainable practices
- Climate resilience
- Responsible AI

It transforms AI from a tool into a trusted decision partner for farmers.

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Thank You